

Online Appendix B: Stochastic Actor-Oriented Models

In addition to our main analytical strategy, we also analyzed the data set with stochastic actor-oriented models (SAOMs), which have been developed to capture dynamic social processes (Snijders, 2011). Detailed explanations can be found in Ripley et al. (2019) and Snijders et al. (2010). These models treat changes in collaboration ties as the result of a continuous-time Markov process in which actors “choose” whom to collaborate with based on an objective function. This actor-oriented approach in principle allows researchers to estimate parameters for theoretically assumed network mechanisms under consideration of network endogenous dynamics.²⁴

A note of caution

Despite the numerous analytical advantages of SAOMs, certain methodological concerns ultimately forced us to abandon SAOMs as part of our primary analytical strategy. One of these concerns was related to a poor goodness of fit, an issue which seems to be impossible to circumvent in applications to large networks at the method’s current developmental stage (Stark, Rambaran, and McFarland 2020: 458, endnote 2; Lewis and Kaufman 2018: 1736–1737). Another problem was that SAOMs assume that actors are aware of all potential partners in the network (Ripley et al., 2019; Snijders et al., 2010). This assumption might be warranted for early stages of neuroblastoma research but it becomes problematic for later

²⁴ It is also possible to account for the longitudinal structure of the data set in the ERGMs framework by controlling for previous collaborations. We performed analyses that controlled for previous collaborations by considering whether dyads had a tie during earlier stages of the network, which is in line with previous studies (e.g., McFarland et al., 2014). These analyses showed very similar results compared with analyses reported in the main text and are available upon request. We decided to report models without controlling for previous collaborations because advancements in the interpretation of longitudinal network models raised doubts about whether an auto-regressive approach is valid for ERGMs (Block et al., 2018). Nevertheless, it is reassuring that our interpretation of the results remains qualitatively unchanged if we enter previous collaborations in our model specifications.

stages as the field quickly grew, encompassing more than 200 researchers from the early 1990s onward.

Therefore, we would like to emphasize that the reported network models should be interpreted only with the greatest caution regarding the substantial claims made in this article. However, we hope that these additional results will be informative for other researchers facing similar problems in their applications of network models to large data sets with changing sizes. Also, changes between models may still provide a qualitative indication of changes in network structure, e.g., if they are compared between periods with similar sizes.

Model specification

We included four parameters in all specifications. The so-called “degree activity effect” captures how popular a researcher is as a collaborator based on her previously accumulated number of coauthorships. A positive estimate indicates that preferential attachment is occurring, and scientists with many coauthor ties maintain and attract more ties over time. The second parameter, “degree homophily” (alias “assortativity”), measures whether authors with a similar number of coauthorships are more or less likely to collaborate. A significantly negative value for this parameter indicates a tendency for authors with many ties to collaborate with others who have fewer coauthorships. The third effect (“Experience of author”) depicts whether authors who have a long experience in the field are more attractive as collaborators: this is the case if the effect turns positive and significant. The fourth parameter is called “Similar experience” and measures whether researchers with similar years of experience are more likely to collaborate with each other than researchers with dissimilar years of experience. We would interpret a significantly positive estimate here as a sign of the presence of status homogeneity in the network. Likewise, we included a term capturing the network popularity of scientists according to their cumulated publications record (“Publications of author”) and a corresponding parameter indicating whether scientists with

similar publication records are more likely to collaborate (“Similar number of publications”). Moreover, we add the “Share of last author positions” term to account for the accumulation of coauthorships that occurs due to being the leader of a research group, i.e., seniority. Likewise, we included a term that captures homogeneity according to seniority (“Similar share of last author positions”).

Control variables

Besides past coauthorship, experience, productivity, and seniority, many other factors are also influential in the formation of scientific collaboration. As highlighted by Wuchty et al. (2007), the average number of coauthor ties increased dramatically over time. To control for the number of ties, we included the “density” term, which is always part of SAOMs and which captures the density of the network. A negative term indicates that fewer ties are present than there would be if a random allocation of ties took place. Previous research illustrated that collaboration is—like many social relationships—bound to foci of activity (Feld, 1991). For instance, coauthorships are more prevalent among researchers in the same department (Dahlander and McFarland, 2013; Stark et al., 2020). To account for these foci of interaction we included a term capturing whether two authors are affiliated with the same institution (“same institution”) or country (“same country”). Moreover, we added the geometrically weighted edgewise-shared partner (GWESP) effect, which models transitivity—the tendency to collaborate with others who collaborate with one’s current coauthors—to further account for the local clustering of ties.

Table B1. Stochastic actor-oriented models SAOMs for collaboration network of neuroblastoma researchers

<i>Periods</i>	1975–1987	1984–1987	1987–1990	1990–1993	1993–1994	1994–1996	1996–1998	1998–2000
Density	-2.56*** (0.41)	-2.83*** (0.24)	14.18 (8.62)	-2.88*** (0.16)	10.15 (11.13)	-3.76*** (0.82)	-2.50*** (0.22)	-3.35*** (0.13)
Geometrically weighted edgewise-shared partners (GWESP)	5.90*** (1.00)	3.60*** (0.25)	10.56** (4.23)	4.10*** (0.20)	12.99 [†] (6.88)	5.09*** (0.46)	4.98*** (0.35)	4.67*** (0.14)
Degree activity (preferential attachment)	0.31*** (0.09)	0.19*** (0.04)	0.80 [†] (0.48)	0.17*** (0.02)	0.87 (0.50)	0.28* (0.15)	0.12*** (0.03)	0.11*** (0.01)
Degree homophily (assortativity)	-0.97*** (0.25)	-0.49*** (0.10)	-3.23 [†] (1.73)	-0.48*** (0.05)	-3.55 (2.20)	-0.75** (0.30)	-0.54*** (0.09)	-0.39*** (0.03)
Same country	-3.03*** (0.67)	-2.36*** (0.35)	-3.40 [†] (1.63)	-2.08*** (0.30)	-2.52 (2.19)	-1.73*** (0.35)	-1.09*** (0.13)	-2.11*** (0.15)
Same institution	1.08 [†] (0.64)	1.41*** (0.37)	-0.61 (1.38)	0.02 (0.26)	-0.62 (1.31)	-0.35 (0.44)	0.02 (0.14)	0.45*** (0.11)
Experience of author	0.02 (0.06)	0.07 [†] (0.037)	0.24* (0.12)	0.06 [†] (0.04)	-0.08 (0.10)	0.14* (0.07)	-0.03 (0.02)	0.01 (0.01)
Similar experience	0.08 (0.56)	0.43 (0.31)	1.66 (1.47)	0.83 (0.60)	-1.22 (1.49)	2.42** (0.80)	0.18 (0.30)	0.12 (0.27)
Publications of author (cumulative)	0.22** (0.08)	0.01 (0.13)	0.35** (0.13)	0.04 (0.04)	0.03 (0.06)	0.08** (0.03)	0.07*** (0.01)	0.03*** (0.01)
Similar number of publications	4.73*** (1.89)	0.42 (0.78)	2.34 (1.51)	0.44 (0.40)	-0.41 (1.56)	2.35*** (0.66)	1.67** (0.64)	2.16*** (0.40)
Share of last author positions	0.15 (0.42)	-0.21 (0.34)	-3.14 (2.32)	0.47* (0.24)	-3.82 (3.20)	-0.16 (0.48)	0.44* (0.21)	0.19 (0.20)
Similar share of last author positions	0.24 (0.39)	-0.26 (0.33)	0.17 (1.10)	0.22 (0.22)	-2.45 (2.65)	-0.27 (0.28)	0.26 (0.21)	0.17 (0.17)
Number of actors	158	152	240	323	364	531	708	721
Overall convergence ratio	0.16	0.17	0.14	0.17	0.15	0.17	0.17	0.14

<i>Periods</i>	2000–2002	2002–2004	2004–2006	2006–2008	2008–2010	2008–2012 ²⁵	2012–2014	2014–2016
Density	-2.09*** (0.29)	-1.37*** (0.21)	-3.59*** (0.11)	-4.93*** (0.14)	-5.83*** (0.42)	-5.97*** (0.18)	-5.70*** (0.74)	-7.96*** (0.51)
Geometrically weighted edgewise- shared partners (GWESP)	5.22*** (0.61)	5.86*** (0.18)	5.00*** (0.14)	4.43*** (0.09)	4.92*** (0.36)	4.99*** (0.12)	4.70*** (0.62)	5.17*** (0.33)
Degree activity (preferential attachment)	0.17*** (0.17)	0.20*** (0.02)	0.10*** (0.01)	0.04*** (0.004)	0.04*** (0.004)	0.05*** (0.003)	0.04*** (0.01)	0.03*** (0.003)
Degree homophily (assortativity)	-0.68*** (0.20)	-0.81*** (0.04)	-0.40*** (0.03)	-0.17*** (0.01)	-0.17*** (0.01)	-0.20*** (0.008)	-0.16*** (0.04)	-0.10*** (0.01)
Same country	-1.60*** (0.18)	-1.89*** (0.12)	-1.70*** (0.09)	-1.51*** (0.06)	-1.28*** (0.10)	-0.98*** (0.05)	-1.36*** (0.19)	-0.61*** (0.04)
Same institution	0.48*** (0.14)	0.26*** (0.09)	0.14* (0.075)	0.26*** (0.06)	0.45*** (0.07)	0.34*** (0.05)	0.42*** (0.05)	0.51*** (0.05)
Experience of author	0.05*** (0.01)	0.01 (0.01)	-0.02* (0.01)	-0.01* (0.005)	-0.007† (0.004)	-0.007* (0.003)	-0.01*** (0.003)	0.01*** (0.003)
Similar experience of authors	1.02*** (0.32)	0.66* (0.28)	0.49* (0.23)	0.11 (0.14)	0.04 (0.10)	-0.02 (0.09)	-0.22 (0.22)	0.28** (0.11)
Publications of author (cumulative)	0.02* (0.01)	0.03*** (0.01)	0.04*** (0.005)	0.02*** (0.002)	0.02*** (0.002)	0.01*** (0.002)	0.02*** (0.002)	0.004*** (0.001)
Similar number of publications	0.87 (0.74)	0.76 (0.71)	3.69*** (0.49)	2.48*** (0.28)	3.02*** (0.24)	2.27*** (0.20)	3.24*** (0.63)	1.69*** (0.23)
Share of last author positions	-0.18 (0.23)	0.10 (0.22)	0.79*** (0.14)	0.44*** (0.11)	0.52*** (0.14)	0.46*** (0.09)	0.45** (0.18)	0.75*** (0.15)
Similar share of last author positions	-0.38 (0.24)	-0.07 (0.19)	0.39*** (0.13)	-0.05 (0.10)	0.15 (0.10)	0.15† (0.08)	-0.002 (0.002)	0.08 (0.17)
Number of actors	814	1155	1257	1334	1517	1850	1585	1417
Overall convergence ratio	0.17	0.23	0.20	0.23	0.20	0.12	0.18	0.20

1481 Note.—Standard errors are reported in parentheses. † $p < 0.10$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$ (two-sided)

²⁵ It was not feasible to estimate a converging model for the period 2010 to 2012. However, defining a period, which entailed three conferences, allowed us to obtain converging estimates for the same model specification as for the other periods.

Results

We now apply SAOMs to investigate how their results relate to our findings reported in the main text.²⁶ We investigated the 18 conferences in 16 periods and ran our model specification separately for each of these periods. We chose this partitioning of the data because it allowed for converging estimates whereas longer periods often proved too complex for the model (cf., Stark et al. 2020: 444). All presented models showed an overall convergence ratio smaller than 0.25, which indicates appropriate convergence (Ripley et al. 2019).

Regarding our theoretical expectation that accumulation dynamics diversify as neuroblastoma research ages, Table B1 shows that parameters capturing the popularity of scientists according to their productivity and seniority did not consistently show statistically significant values until the second half of the 1990s. In a similar vein to our results in the main text, SAOMs indicated that productivity became a relevant factor for scientists' popularity as coauthors from 1994 onwards. It took even longer for seniority to contribute to scientists' accumulation of coauthorships (from 2004 onward). In comparison, preferential attachment was present for almost all periods.

Concerning our expectation that status homogeneity becomes stronger as a field matures, SAOMs produced results that were broadly consistent with our main analyses: as the field entered the second half of its development, the parameter capturing whether scientists with a similar publication record are more likely to collaborate gained in size and was consistently significant after 2004. In line with our conjecture that scientists with many coauthor ties tend to collaborate with others holding fewer ties because they have just entered the field as PhD students or postdoctoral researchers, we observed a consistently negative significant

²⁶ Given that the networks under investigation are undirected, we followed the advice provided in Ripley et al. (2019: 52) to choose a model type informing the estimation procedure. We chose model type 3, which assumes that one actor takes the initiative in proposing a new or resolving an existing tie. While it is only possible for a new tie to form if the other actor agrees, no consent of the other actor is needed to resolve a tie.

parameter for degree heterophily.²⁷ This corroborates the notion that while status-similar collaborations should become more prevalent as a stratified order emerges, the mentor-apprentice model of collaboration should not lose its importance for the organization of research activity.

We would like to note that the observed trends also hold for the time span from 2002 onwards. Periods in this part of the network's development show sizes between 1,155 and 1,585 actors.²⁸ While this is still a relevant shift in network size, previous accounts compare estimates from models that differ similarly or even more strongly in network size (An, 2015; Goodreau et al., 2009; Kronegger et al., 2012; McFarland et al., 2014; Simpson, 2019; Stark et al., 2020). Therefore, we interpret our results as at least a qualitative indication of changes in network structure. However, future research should address the problem of comparing longitudinal network models between periods with starkly differing sizes and re-evaluate our results and interpretations.

Relative importance measures

While we limit our discussion of SAOM results to a heuristic comparison of coefficients over time, we also considered the measures of relative importance (RI) proposed by Indlekofer and Brandes (2013). In principle, these measures allow a comparison of the relative importance of different network tendencies for actors' collaboration choices (for recent applications, see

²⁷ This finding could also partially be due to the “friendship paradox” discovered by Feld (1991). Feld showed that the average number of friendships one's friends have is almost always higher than one's own number of friendships. In essence, the disproportionate weighting of friends with many ties in the calculation of one's friends' average number of friendships is responsible for this phenomenon. From this, it follows that scholars probably have fewer coauthorship ties than their coauthors have (on average.)

²⁸ Please note that the number of actors for the period 2008–2012 is not comparable because this period encompasses three instead of two conferences. The reason for this decision was an issue with model degeneracy; please see the corresponding footnote in Table B1.

Rambaran et al., 2020; Schaefer and Kreager, 2020; Stark et al., 2020). However, we decided not to use RIs for several reasons. First, these measures do not consider uncertainty in estimates—many coefficients were large but insignificant in early periods—making the interpretation of their RIs problematic. Second, RIs are an aggregation of individual scores and actors in our data set exhibit strong heterogeneity in terms of overall degrees and the distribution of attributes. While this is not a problem per se, it strains credulity as to how meaningful RIs are in our case. Third, the current implementation of RIs only covers undirected networks of type 2, meaning that one actor can propose or dissolve a tie without the other actor's consent. Instead, though, we chose model type 3, which assumes that one actor takes the initiative in proposing or resolving a tie, but the other actor has to agree if a tie is initiated (Ripley et al., 2019).